

NASA JOHN C. STENNIS SPACE CENTER
(SSC) CENTER OPERATIONS DIRECTORATE
(COD) STATEMENT OF WORK (SOW)

Medium Voltage Transformer – WFTCA-1C Electrical Power Upgrades at HPGF

1. PURPOSE AND BACKGROUND

This Statement of Work (SOW) defines the scope, technical requirements, deliverables, and acceptance criteria for the purchase and delivery of one (1) medium voltage (MV) pad-mounted transformer in support of Project WFTCA-1C – Electrical Power Upgrades at the High-Pressure Gas Facility (HPGF) within the SSC Test Complex. The transformer is a long-lead, critical-path equipment item required to maintain the program schedule for HPGF electrical upgrades and related integration activities.

2. SCOPE OF SUPPLY

The Contractor shall furnish, test, and deliver a fully compliant transformer assembly suitable for pad-mount installation at HPGF, including factory tests, documentation, and commissioning support plans as specified herein. The scope is exclusively for equipment; installation and construction services are out of scope.

3. TECHNICAL REQUIREMENTS

3.1 Nameplate Ratings and Electrical Characteristics

- a. Primary: 15 kV class with taps (ANSI standard tap range to be proposed to meet utility/ feed variation).
- b. Secondary: 4160 V (3-phase)
- c. Phase: Three-phase.
- d. Capacity: 5,000 kVA.
- e. Frequency: 60 Hz.

3.2 The transformer shall comply with the following applicable codes and standards:

- a. IEEE 386 (2016) Separable insulated connector system for power distribution systems rated 2.5 kV through 35 kV.
- b. IEEE C2 (2023) National Electrical Safety Code.
- c. IEEE C37.42 (2016) Specification for high-voltage (>1,000 V) fuses and accessories.
- d. IEEE C57.12.00 (2021) General requirements for liquid-immersed distribution, power, and regulating transformers.

- e. IEEE C57.12.28 (2023) Standard for pad-mounted equipment – enclosure integrity.
- f. IEEE C57.12.34 (2022) Standard requirements for pad-mounted, compartmental-type, self-cooled, three-phase distribution transformers (10 MVA and smaller; 34.5 kV and below).
- g. IEEE C57.12.80 (2010) Standard terminology for power and distribution transformers.
- h. IEEE C57.12.90 (2021) Test code for liquid-immersed distribution, power, and regulating transformers.

3.3 Construction and Materials

- a. Tank: Stainless steel (pad-mounted configuration).
- b. Windings: Copper.
- c. Dielectric: Mineral oil per applicable ASTM standards; provide oil data sheet and certificate of analysis.
- d. Enclosure: Outdoor, pad-mounted, tamper-resistant per applicable IEEE/ANSI/NEMA standards.
- e. Bushings, tap changers, and accessories appropriate to ratings above; Contractor to provide a complete bill of materials.

3.4 Design Standards and Quality System

- a. The Contractor shall maintain and apply a quality system compliant with ISO 9001 or equivalent higher-level quality standard applicable to COTS electrical equipment.
- b. Equipment shall be delivered as commercial off-the-shelf (COTS)

3.5 Factory Testing

- a. Perform standard factory tests for MV power transformers (e.g., ratio, polarity, excitation current, insulation resistance, dielectric tests, routine oil quality tests).
- b. Provide a Factory Acceptance Test (FAT) plan and test reports prior to shipment.

3.6 Commissioning Support Plan

- a. Provide a commissioning plan, including recommended installation checks, energization procedures, and acceptance criteria aligned with OEM recommendations.

4. DELIVERABLES

4.1 Equipment

a. One (1) MV transformer meeting Sections 3.1–3.3.

4.2 Documentation

- a. As-built equipment drawings, nameplate data sheets, and full test reports (FAT).
- b. Oil certificate of analysis and material certifications (Certificate of Conformance).
- c. Commissioning plan and maintenance manual.
- d. Warranty certificate and terms.
- e. Confirmation the transformer meets all IEEE standards listed within the SOW.

5. SCHEDULE AND DELIVERY

5.1 Required Delivery Date

a. Required on-site no later than May 2027 (or earlier), reflecting long-lead fabrication duration. The current independent estimate indicates a manufacturing lead time of approximately 53 weeks; offeror shall propose a committed lead/delivery schedule not exceeding program needs.

5.2 Delivery Location

a. NASA John C. Stennis Space Center (SSC), High Pressure Gas Facility (HPGF), Test Complex, RA40 receiving (final site coordinates and delivery gate instructions will be provided at award).

5.3 Packaging and Shipping

- a. Ship FOB Destination, SSC; provide shock indicators, moisture protection, and handling instructions suitable for MV transformer transport.
- b. Provide lifting points and rigging guidance; note that installation services are not included in this SOW.

6. INSPECTION AND ACCEPTANCE

6.1 Upon delivery to SSC, Government acceptance testing/inspection will be performed within 30 days to ensure suitability.

6.2 Acceptance is contingent upon satisfactory review of documentation, visual inspection, and any incoming tests specified in the commissioning plan.

7. SAFETY, HEALTH, AND ENVIRONMENTAL

7.1 Safety & Health

- a. The transformer procurement does not include hazardous or potentially hazardous

devices requiring special Center SMA approvals; installation-related lifts/rigging, if any, are managed under separate scope.

7.2 NEPA Compliance

a. The activity is anticipated to fall under NEPA CatEx 1(i) with no extraordinary circumstances; SSC environmental review will be completed as required.

8. PROPERTY MANAGEMENT

8.1 Tagging and Records

a. Contractor shall provide serial numbers and nameplate data to facilitate Government property records.

9. WARRANTY

9.1 Provide a minimum 24-month warranty from final acceptance, covering defects in materials and workmanship; include on-site technical support terms for failure analysis if required.

10. INTERFACES AND COORDINATION

10.1 The Contractor shall coordinate with the SSC Receiving Office for delivery scheduling and site access and shall provide advance notice of shipment at least 10 business days prior to arrival.

10.2 All technical requirement questions post-award shall be addressed to the SSC project technical authority (RA40/COD).

11. SUBMITTALS

11.1 Pre-Shipment Submittals

- a. FAT plan and test procedures.
- b. Final drawings, datasheets, and QA certifications (ISO 9001 conformity statement).

11.2 Post-Shipment Submittals

- a. Completed FAT reports, oil certificate of analysis, and commissioning plan.

— END OF STATEMENT OF WORK —